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Data boundary services for personal wearable system in intelligence textile networks

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Abstract

Data boundary analysis is an important issue in textile industry. Traffic converges in the converged nodes which responsible for the capacity calculation of data boundary. It includes the maximum boundary and the corresponding minimum boundary. At present, the research in this field is not detailed enough. However, its importance is self-evident. If there is no data boundary as support, the configuration and operation of the system will become very redundant. In order to obtain more concise system configuration, the probability of boundary and system model are studied. Finally, the accuracy of the model is simulated and compared with numerical simulation.

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1. Introduction

Intelligence textile networks was activated by countless emergencies. With the help of data boundary calculation, it can make the intelligence textile configuration indicators more certain. This will gain an advantage in design and cost of sales. Traffic converges in the converged nodes which responsible for the capacity calculation of data boundary. It includes the maximum boundary and the corresponding minimum boundary. At present, the research in this field is not detailed enough. However, its importance is self-evident. If there is no data boundary as support, the configuration and operation of the system will become very redundant. In order to obtain more concise system configuration, the probability of boundary and system model are studied.

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We assume that one of the important nodes 10 sends data stream continuously. Nodes 11 continuously forward the data stream sent by the source node. Node 21 is the data fusion calculation of multi-channel forwarding node, in order to obtain more computing resources. Node 31 is to summarize and analyze the data streams from different sources, as shown in Figure 1.

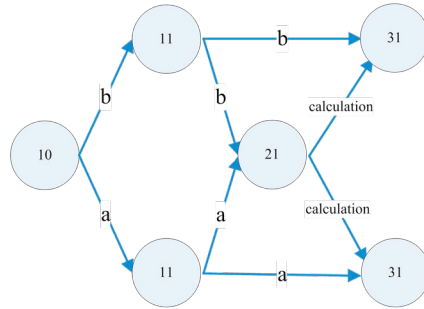


Fig. 1. Data fusion calculation flow chart.

2. Data fusion calculation

The input stream of intelligence textile networks is defined as the emergency flow. It includes a variety of mathematical models, whose distribution is determined by user arrival events and service cycles.

2.1. Variable

We have a number of different variable symbols. For example, upper and lower boundaries and widths.

$$A = \{a(\bullet) : \forall 0 \leq u1 \leq u2, 0 \leq a(u1) \leq a(u2)\} \quad (1)$$

At the same time, its anti boundary function is defined.

$$\overline{A} = \{a(\bullet) : \forall 0 \leq u1 \leq u2, 0 \leq a(u2) \leq a(u1)\} \quad (2)$$

The service functions mentioned in the following paper all satisfy this relation.

2.2. Concept definition of calculation and analysis

The convolution function calculation is

$$(a \otimes b)(t) = \inf_{0 \leq s \leq t} (a(s) + b(t-s)) \quad (3)$$

The calculation of negative convolution function is

$$(a - b)(t) = \inf_{0 \leq s \leq t} (a(s) + b(t-s)) \quad (4)$$

Another definition of algebraic function calculation

$$(a \vee b)(t) = \inf_{0 \leq s \leq t} (a(s) + b(t-s)) \quad (5)$$

3. Data service model calculation in textile networks

If the data reaches the cumulative model, it obeys two boundary requirements.

$$Por\{Q(u_x, u_y) - \alpha(u_y - u_x) > a\} \leq A(u_x) \quad (6)$$

and

$$Por\{O(u_x, u_y) < \beta(u_y - u_x) - a\} \leq B(u_x) \quad (7)$$

4. Analysis of processing capacity of node model

The fusion boundary conditions input by multiple data streams can be defined as:

$$A(u1) = (a_1 \otimes \dots \otimes a_n)(u1), \quad \alpha(u2) = \vee_{i=1}^n \alpha_i(u2)$$

It can be represented in Figure 2.

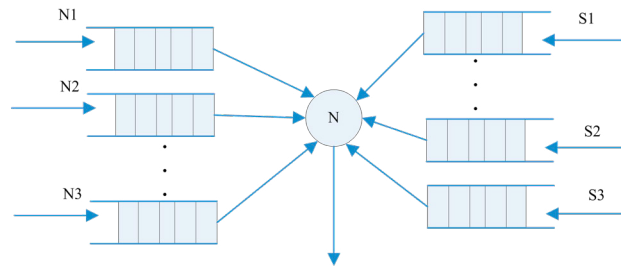


Fig. 2. Processing capacity of node model.

$$\begin{aligned} & Por(Q(u1, u2) - e(u2 - u1) > a) \\ &= \sum_{1 \text{ and } 2} Por(\{Q_1(u1, u2) - \alpha_1(u2 - u1)\} + \{A_2(u1, u2) - \alpha_2(u2 - u1)\} > a) \\ &\leq \inf_{z < x} (Por\{Q_1(u1, u2) - \alpha_1(u2 - u1)\} > K \\ &\quad + \{Q_2(u1, u2) - \alpha_2(u2 - u1)\} > a - K) \\ &\leq \inf_{z < x} (A_1(K) + A_2(a - K)) \\ &= \sum_{< 2} (A_1 \otimes A_2)(a) \end{aligned} \quad (8)$$

The more common case is as follows.

$$\begin{aligned}
& Por(Q(u1, u2) - e(u2 - u1) - e(u3 - u2) - e(u4 - u3) > a) \\
&= \sum_{>2} Por(\{Q_1(u1, u2) - \alpha_1(u2 - u1)\} + \{A_2(u1, u2) - \alpha_2(u2 - u1)\} > a) \\
&\leq \inf_{z < x} (Por\{Q_1(u1, u2) - \alpha_1(u2 - u1)\} > K \\
&\quad + \{Q_2(u1, u2) - \alpha_2(u2 - u1)\} > a - K) \\
&\leq \inf_{z < x} (A_1(K) + A_2(a - K)) \\
&= \sum_{>2} (A_1 \otimes A_2)(a) \\
&\leq (A_1 \otimes \dots \otimes A_N)(a)
\end{aligned} \tag{9}$$

5. Node processing

The node data flow after fusion can be expressed as

$$\begin{aligned}
& Por(L(u1, u2) > a) \\
&= \sum_{>2} Por(\{L_1(u1, u2) - \alpha_1(u2 - u1)\} + \{L_2(u1, u2) - \alpha_2(u2 - u1)\} > a) \\
&\leq \inf_{z < x} (Por\{A_1(u1, u2) - \alpha_1(u2 - u1)\} > K \\
&\quad \otimes \{B_2(u1, u2) - \alpha_2(u2 - u1)\} > a - K) \\
&= \sum_{>2} (A_1 \otimes A_2)(a) \\
&\leq (A_1 \otimes L \otimes L \otimes A_N)(a) \\
&= \sum_{>2} \alpha_i(u1, u2)
\end{aligned} \tag{10}$$

Proof:

For $i \in (1, 2, \dots, n)$

$$\begin{aligned}
& Por(L1(u1, u2) + L2(u1, u2) > a) \\
&= \sum_{L1 \text{ and } L2 > 2} Por(\{L_1(u1, u2) - \alpha_1(u2 - u1)\} + \{L_2(u1, u2) - \alpha_2(u2 - u1)\} > a) \\
&\leq \inf_{z < x} (Por\{A_1(u1, u2) - \alpha_1(u2 - u1)\} > K \\
&\quad \otimes \{B_2(u1, u2) - \alpha_2(u2 - u1)\} > a - K) \\
&= \sum_{>2} (A_1 \otimes A_2)(a) \\
&\leq (A_1 \otimes L \otimes L \otimes A_N)(a) \\
&= \inf_{z < x} Por\{L_1(u1, u2) > b + PorL_1(u2 - u1) > a - b \\
&= A \otimes B(a)
\end{aligned} \tag{11}$$

Therefore, a further expression of P can be obtained

$$\begin{aligned}
& \sum_{L1 \text{ and } L2 > 2} \text{Por}(\{B_i(u_x, u_y) - \alpha_i(u_y - u_x)\} + \{L_i(u_x, u_y) - \alpha_i(u_y - u_x)\} > a) \\
& \in (\text{Por}\{A_1(u_x, u_y) - \alpha_1(u_y - u_x)\} > K \\
& \quad \otimes \{B_2(u_x, u_y) - \alpha_2(u_y - u_x)\} > a - K) \\
& = \sum_{> 2} (A_1 \otimes A_2)(a) \\
& \leq (A_1 \otimes L \otimes L \otimes A_N)(a) \\
& = \inf_{z < x} \text{Por}\{L_1(u_x, u_y) > b + \text{Por}L_1(u_y - u_x) > a - b \\
& = (0, \alpha_i(u_x, u_y) - B_i(u_y - u_x))
\end{aligned} \tag{12}$$

For the proof under a certain interval, it can be expressed as

$$\begin{aligned}
& \sum_{L1 \text{ and } L2 > 2} \text{Por}(\{L_i(u_x, u_y) - \alpha_i(u_y - u_x)\} + \{L_i(u_x, u_y) - \sum \alpha_i(u_y - u_x)\} > a) \\
& < (\{A_1(u_x, u_y) + B(a + b(u_y - u_x)) - \sum \alpha_1(u_y - u_x)\} \\
& \text{and } \{B_2(u_x, u_y) - \alpha_2(u_y - u_x)\} > a - K)
\end{aligned} \tag{13}$$

Proof:

$$\begin{aligned}
& \sum_{L1 \text{ and } L2 > 2} \text{Por}(\{L_i(u_x, u_y) - \alpha_i(u_y - u_x)\} + \{L_i(u_x, u_y) - \alpha_i(u_y - u_x)\} > a) \\
& = (\{A_1(u_x, u_y) + B(a + b(u_y - u_x)) - \alpha_1(u_y - u_x)\} \\
& \text{and } \{B_2(u_x, u_y) - \alpha_2(u_y - u_x)\} > a - K) \\
& = a + \sum_{L1 \text{ and } L2 > 2} \text{Por}(B(u_y - u_x) - A(u_y - u_x)) \\
& (A1(u_x, u_y) + A2(u_x, u_y) + A3(u_x, u_y)) \\
& = (A + B) \sum_{L1 \text{ and } L2 > 2} \text{Por}((K(u_x, u_y) - a(u_x, u_y))
\end{aligned} \tag{14}$$

6. Conclusions

This paper studies the data fusion problem in general networks. The algebraic method is used to prove the existence of data boundary and how to obtain the optimal boundary. Based on the basic two-dimensional vector proof, the difficulty of multi-dimensional data stream fusion is proved. How to select the system configuration under different data acquisition modes is given in intelligence textile networks.

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